

AI ASSISTED LAB ASSIGNMENT – 2.1

Exploring Additional AI Coding Tools beyond Copilot

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Task-1: Statistical Summary for Survey Data

Input:

#generate a Python function that reads a list of numbers and calculates the mean, minimum, and maximum values.

```
def calculate_statistics(numbers):
```

```
    if not numbers:
```

```
        return None, None, None # Return None for mean, min,
and max if the list is empty
```

```
    mean_value = sum(numbers) / len(numbers)
```

```
min_value = min(numbers)
```

```
max_value = max(numbers)
```

```
return mean_value, min_value, max_value
```

```
#example usage:
```

```
numbers = [10, 20, 30, 40, 50]
```

```
mean, minimum, maximum = calculate_statistics(numbers)
```

```
print(f"Mean: {mean}, Minimum: {minimum}, Maximum:  
{maximum}")
```

Explanation:

The `calculate_statistics()` function efficiently calculates the **mean**, **minimum**, and **maximum** of a list of numbers. It first checks whether the list is empty to avoid errors. Then it uses `sum()` and `len()` to calculate the mean and the built-in `min()` and `max()` functions to find the smallest and largest values.

Output:

```
PS C:\Users\Prasad reddy\.vscode> & "C:\Users\Prasad reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad reddy/.vscode/python.py/lab.py"  
● Mean: 30.0, Minimum: 10, Maximum: 50  
PS C:\Users\Prasad reddy\.vscode> |
```

Task-2: Armstrong Number – AI Comparison

Input:

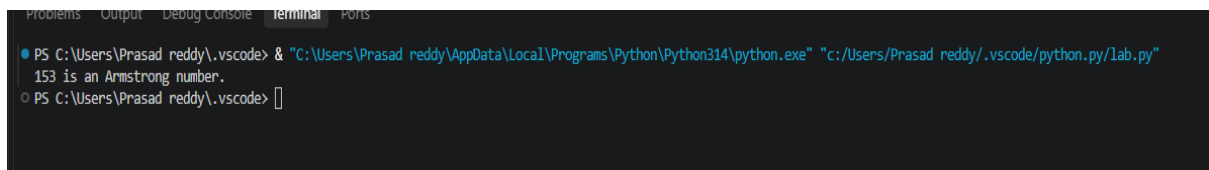
#Generate an Armstrong number checker using Given a number, check if it is an Armstrong number or not. Compare their outputs, logic style, and clarity.

```
def is_armstrong_number(n):  
    # Convert the number to a string to easily iterate over each digit  
    digits = str(n)  
    num_digits = len(digits)  
  
    # Calculate the sum of each digit raised to the power of the number of digits  
    armstrong_sum = sum(int(digit) ** num_digits for digit in digits)  
  
    # Check if the calculated sum is equal to the original number  
    return armstrong_sum == n  
  
# Example usage:  
number = 153  
if is_armstrong_number(number):  
    print(f"{number} is an Armstrong number.")  
else:  
    print(f"{number} is not an Armstrong number.")
```

Explanation:

An Armstrong number is a number in which the sum of each digit raised to the power of the number of digits is equal to the original number.

Output:



```
PS C:\Users\Prasad_reddy\.vscode> & "C:\Users\Prasad_reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad_reddy/.vscode/python.py/lab.py"
153 is an Armstrong number.
PS C:\Users\Prasad_reddy\.vscode>
```

Task-3: Leap Year Validation Using Cursor AI

Input:

#generate a Python program that checks whether a given year is a leap year.also give two different methods to check for leap year and compare their outputs, logic style, and clarity.

```
def is_leap_year_method1(year):
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
        return True
    else:
        return False

# Example usage:
year = 2020

if is_leap_year_method1(year):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")
```

Method 2: Using a more concise approach

```
def is_leap_year_method2(year):  
    return year % 4 == 0 and (year % 100 != 0 or year % 400 == 0)
```

Example usage:

```
year = 2021
```

```
if is_leap_year_method2(year):
```

```
    print(f"{year} is a leap year.")
```

```
else:
```

```
    print(f"{year} is not a leap year.")
```

#example usage:

```
year = 2024
```

```
if is_leap_year_method1(year) == is_leap_year_method2(year):
```

```
    print(f"Both methods agree that {year} is a leap year:  
{is_leap_year_method1(year)}")
```

Explanation:

A leap year is a year that contains 366 days instead of 365. In Python, we can check whether a year is a leap year using divisibility rules. A year is a leap year if it is divisible by 4, but century years must also be divisible by 400. The program uses two methods to perform this check. The first method uses an if-else statement, while the second method uses a shorter Boolean expression. Both methods produce the same output. Method 1 is easier for beginners to understand, while Method 2 is more concise and clear for experienced programmers.

Output:

```
PS C:\Users\Prasad reddy\.vscode> & "C:\Users\Prasad reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad reddy/.vscode/python.py/lab.py"  
2020 is a leap year.  
2021 is not a leap year.  
Both methods agree that 2024 is a leap year: True  
PS C:\Users\Prasad reddy\.vscode> |
```

Task-4: Student Logic + AI Refactoring (Odd/Even Sum)

Input:

#Write a Python program that calculates the sum of odd and even numbers in a tuple.

#Write a Python program that calculates the sum of odd and even numbers in a tuple,

```
def sum_odd_even(numbers):  
  
    odd_sum = sum(x for x in numbers if x % 2 != 0)  
  
    even_sum = sum(x for x in numbers if x % 2 == 0)  
  
    return odd_sum, even_sum
```

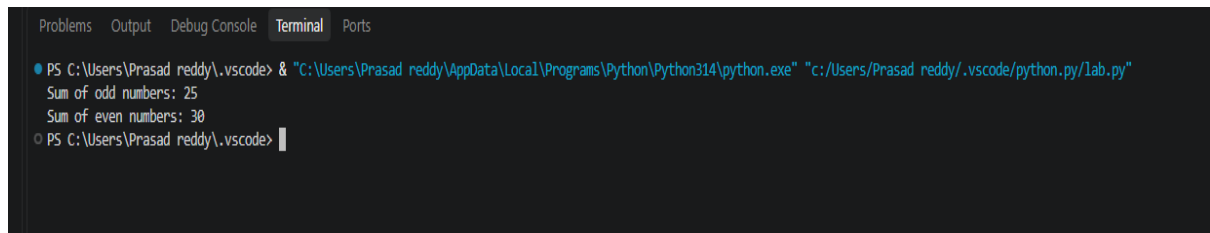
Example usage:

```
tuple_numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10)  
  
odd_total, even_total = sum_odd_even(tuple_numbers)  
  
print(f"Sum of odd numbers: {odd_total}")  
  
print(f"Sum of even numbers: {even_total}")
```

Explanation:

This Python program calculates the **sum of odd and even numbers** in a tuple. The function checks each number using the modulus (%) operator. If the remainder after division by 2 is not zero, the number is odd; otherwise, it is even. The sum() function adds the numbers separately. Finally, the program displays the total of odd numbers as **25** and even numbers as **30**.

Output:



The image shows a screenshot of a Visual Studio Code terminal window. The terminal has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The command prompt shows the execution of a Python script located at 'c:/Users/Prasad reddy/.vscode/python.py/lab.py'. The script outputs two lines: 'Sum of odd numbers: 25' and 'Sum of even numbers: 30'. The prompt then returns to the command line.

```
PS C:\Users\Prasad reddy\.vscode> & "C:\Users\Prasad reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad reddy/.vscode/python.py/lab.py"
Sum of odd numbers: 25
Sum of even numbers: 30
PS C:\Users\Prasad reddy\.vscode>
```